Class Activity # 2
Calculus Physics 2

ame: Key Date: _____

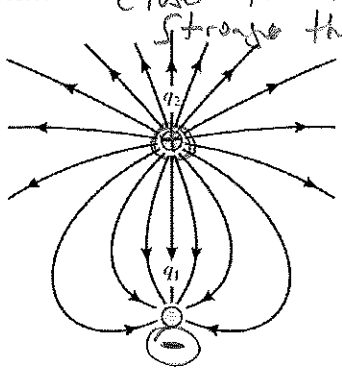
Group Members: _____ _____ _____

Solve the following problems together as a group, stay engaged in the conversation with the group!

- Use **Lecture Notes Chapter 36 Parts A, B and C through 37** as a guide to answer the following questions.

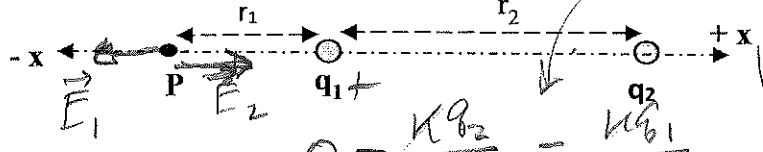
1) Two stationary point charges q_1 and q_2 are shown in the figure along with a sketch of some field lines representing the electric field produced by them. What can you deduce from the sketch?

Closer the lines stronger the field



A) q_1 is negative and q_2 is positive; the magnitudes are equal.
B) q_1 and q_2 have the same sign; the magnitudes are equal.
C) q_1 is positive and q_2 is negative; the magnitude of q_1 is greater than the magnitude of q_2.
(D) q_1 is negative and q_2 is positive; the magnitude of q_1 is less than the magnitude of q_2. $\vec{E}$ - field goes from + → −
E) q_1 and q_2 have the same sign; the magnitude of q_1 is greater than the magnitude of q_2.

2) Two point charges are shown below, what must the magnitude of the net charge on q_2 and type, to create a net electric field equal to zero at point P. The net charge on q_1 = 20 nC and is positive, r_1 = 0.5 m and r_2 = 1.5 m.



$\Sigma \vec{E} = \vec{E}_1 + \vec{E}_2 = 0$

$\Sigma \vec{E} = (|\vec{E}_2| - |\vec{E}_1|)\hat{x}$

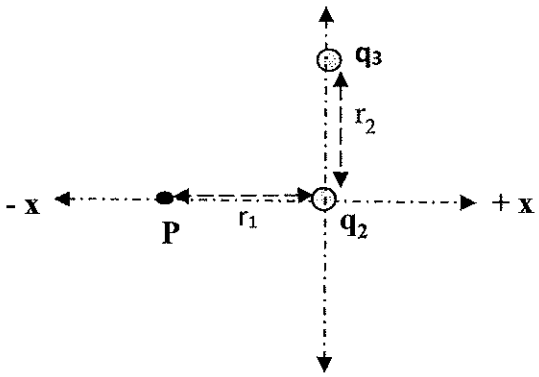
$0 = \frac{kq_2}{(r_1+r_2)^2} - \frac{kq_1}{r_1^2}$

$q_2 = \frac{(r_1+r_2)^2}{r^2} q_1 = 3.2 \times 10^{-7} C$

3) What is the magnitude and direction of the net Electric field strength at point P due to the point charges q_2 and q_3? ($k = 9.0 \times 10^9\ Nm^2/C^2$)

q_2 = 50 nC negative
q_3 = 50 nC negative
r_1 = 0.8 m
r_2 = 0.8 m

See attached

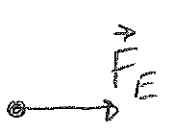


$|\Sigma \vec{E}_p| = 984\ N/C$

$\alpha = 15°$

$\beta = 75°$

4) A small object with a 5 μC charge is accelerating horizontally on a frictionless surface at 0.0050 m/s² due only to a uniform electric field. If the object has a mass of 0.02 kg, what is the magnitude of the electric field?



$\vec{F}_E$

$\Sigma \vec{F} = \vec{F}_E = m\vec{a}$

$\Sigma F_x = |\vec{F}_E| = m a_x$

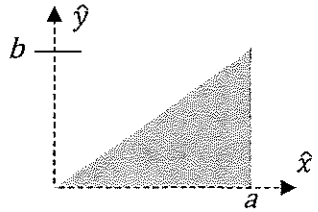
$q|\vec{E}| = |\vec{F}_E| = m a_x$

Solve for $|\vec{E}|$

$|\vec{E}| = \frac{m a_x}{q} = 20\ N/C$

5) Determine the total surface charge on a triangular shaped area shown below in terms of a, b and c, the length of the base is given as "a" and the height given as "b". The surface charge density varies along the x-axis as $\sigma = cx^2$, where c is a constant.

See attached



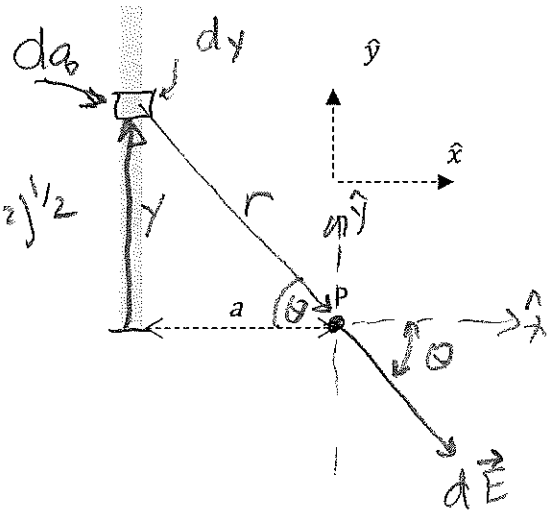
6) A positively charged straight wire with a uniform linear charge density of λ, determine the electric field strength at point P, "a" meters above the wire and situated at the bottom of the wire of length L. Assume that a and L are constant and do not change in this problem.

A) Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.

B) Determine the incremental charge on the wire.

$dq = \lambda\, dL = \lambda\, dy$

$r = \sqrt{y^2 + a^2}$ $\quad \cos(\theta) = \frac{a}{r} = \frac{a}{(y^2+a^2)^{1/2}}$



C) Determine the electric field strength along the x-axis.

See attached

D) Determine the electric field strength along the y-axis.

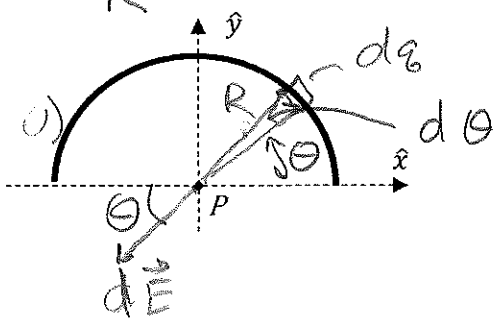
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7) Determine the electric field strength at point P at the center of a semi-circular piece of wire with a uniform charge density λ.

$d\theta = \frac{dL}{R} \rightarrow dL = R\,d\theta$

A) Electric field strength along the x-axis.

$dq = \lambda\, dL = \lambda R\, d\theta$



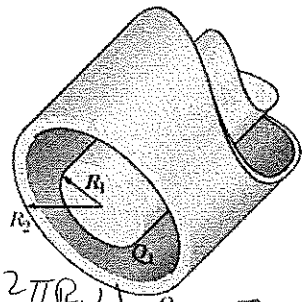
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B) Electric field strength along the y-axis.

See attached

$\vec{E} = (0\ N/C)\hat{x} + \left(-\frac{2k\lambda}{R}\right)\hat{y}$

8) The cross-section of a coaxial cable is shown below; it consists of an inter conductor with a radius of $R_1 = 1.5$ mm and a length of 2 m. The surface charge density of the enter cable is a positive $\sigma_1 = 4 \times 10^{-10}$ C/m^2. We will assume that air/vacuum fills the space between the inter conductor and outer conductor (in practice this is not the case). The outer conductor has an outside radius of $R_2 = 15$ mm and with the same length of 2 m the same as the inter conductor. The surface charge density on the outside surface of the outer cable is a negative $\sigma_2 = 8 \times 10^{-12}$ C/m^2. The outer conductor has a thickness of 1 mm. Using Gauss's Law answer the following questions.



A) Deterine the total charge on the inter conductor (you can ignore the area of the ends of the cable).

Area of Cylinder no ends

$A = 2\pi R_1 L$

$Q_T = \sigma_1 A = \sigma_1 (2\pi R_1 L)$ $Q_{in} = 7.54 \times 10^{-12}$ C positive

B) Determine the amount of charge and type on the inter surface of the outside conductor.

Inter Conductor has $Q_{in} = 7.54 \times 10^{-12}$ C positive

This would induce a charge on the inside surface of the Outer Conductor of 7.54×10^{-12} C negative

C) Determine the magnitude of the electric field strength in the space between the two conductors at a radius of r = 12 mm.

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0}$

$|\vec{E}|A = \frac{Q_{in}}{\varepsilon_0}$

$Q_{in} = \sigma_1 (2\pi R_1 L)$

Area of Gaussian Surface

$A_G = 2\pi r L$

$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{\sigma_1 2\pi R_1 L}{\varepsilon_0 2\pi r L}$

$|\vec{E}| = \frac{\sigma_1 R_1}{\varepsilon_0 r} = 5.65\ N/C$

need $Q_{in} = Q_1 + Q_2 = +\sigma_1 2\pi L R_1 + (-\sigma 2\pi L R_2) = ?$

$= 2\pi L(\sigma_1 R_1 - \sigma_2 R_2)$

D) Determine the electric field strength outside the cable at a distance of r = 30 mm.

$$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A_G} = \frac{2\pi L(\sigma_1 R_1 - \sigma_2 R_2)}{\varepsilon_0 (2\pi L r)} = \frac{\sigma_1 R_1 - \sigma_2 R_2}{r \varepsilon_0} = 1.8\ N/C$$

9) A solid insulating sphere has a volume charge density ρ and a radius R.

A) Determine the equation for the electric field strength inside the sphere at a distance r_{in} from the center. ($r_{in} < R$).

$Q_{in} = \rho V = \rho\left(\frac{4}{3}\pi r_{in}^3\right)$

$A_G = 4\pi r_{in}^2$

$$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A_G} = \frac{\rho\left(\frac{4}{3}\pi r_{in}^3\right)}{\varepsilon_0 4\pi r_{in}^2}$$

$$|\vec{E}| = \frac{\rho r_{in}}{3\varepsilon_0}$$

B) Determine the equation for the electric field strength outside the sphere at a distance r_{out} from the center. ($r_{out} < R$).

$Q_{in} = \rho V = \rho\left(\frac{4}{3}\pi R^3\right)$

$A_G = 4\pi r_{out}^2$

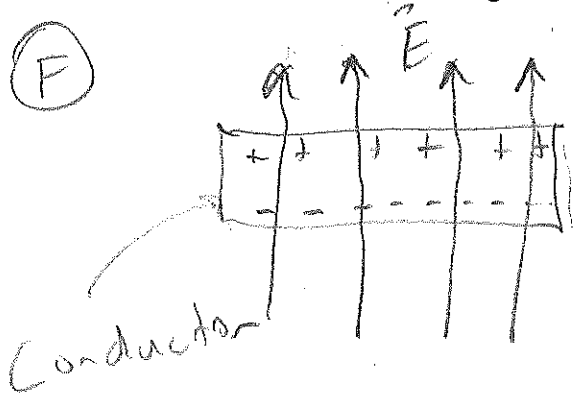
$$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{\rho\left(\frac{4}{3}\pi R^3\right)}{\varepsilon_0 (4\pi r_{out}^2)}$$

$$|\vec{E}| = \frac{\rho R^3}{3\varepsilon_0 r_{out}^2}$$

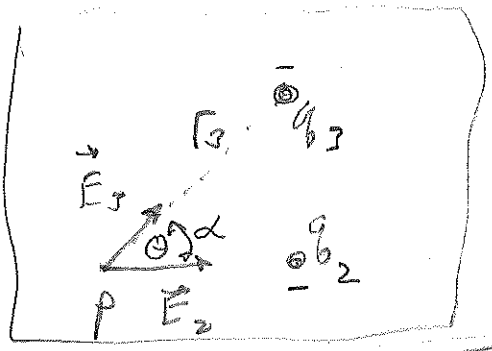
10) An electrically neutral, hollow conducting aluminum cube rests on a wooden table in a region where a uniform external electric field of 100 N/C is directed straight upward. Which statement is true concerning the aluminum cube? (There can be more than one true statement).

(A) (F)

(A) The top surface is charged positively.
B) The top surface is charged negatively.
C) The top surface is neutral.
D) The field inside the cube is 100 N/C.
E) The field inside the cube is greater than 100 N/C.
(F) The field inside the cube is 0 N/C.



3)



$\Sigma \vec{E} = \vec{E}_2 + \vec{E}_3$

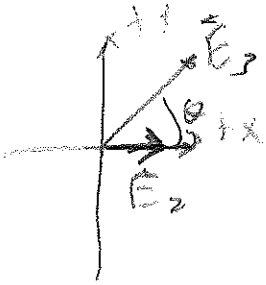
$\Sigma \vec{E} = (\Sigma E_x)\hat{x} + (\Sigma E_y)\hat{y}$

$r_3 = \sqrt{r_1^2 + r_2^2} = 1.13\,m$

$\theta = \tan^{-1}\left(\frac{r_2}{r_1}\right) = 45°$

$|\vec{E}_2| = \frac{Kq_2}{r_2^2} = 703\,N/C$

$|\vec{E}_3| = \frac{Kq_3}{r_3^2} = 352\,N/C$



$\Sigma E_x = \Sigma \vec{E} \cdot \hat{x} = |\vec{E}_2|\cos(\alpha_2) + |\vec{E}_3|\cos(\alpha_3)$

$\alpha_2 = 0°$ $\quad \alpha_3 = 45°$

$\Sigma E_x = 703\,N/C + 249\,N/C = 952\,N/C$

$\Sigma E_y = \Sigma \vec{E} \cdot \hat{y} = |\vec{E}_2|\cos(\beta_2) + |\vec{E}_3|\cos(\beta_3)$

$\beta_2 = 90°$ $\quad \beta_3 = 90° - 45° = 45°$

$\Sigma E_y = 0 + 249\,N/C = 249\,N/C$

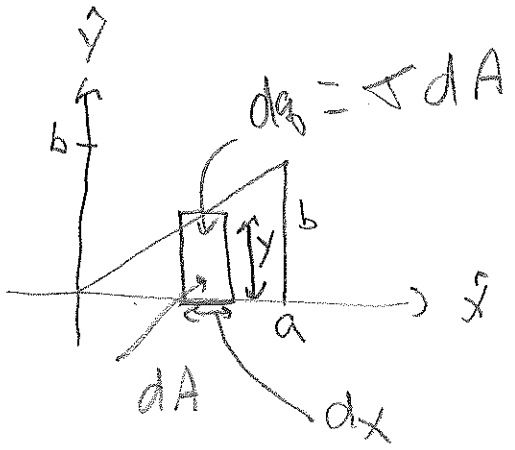
$\Sigma \vec{E} = (952\,N/C)\hat{x} + (249\,N/C)\hat{y}$

$|\Sigma \vec{E}| = \sqrt{E_x^2 + E_y^2} = 984\,N/C$

$\alpha = \cos^{-1}\left(\frac{E_x}{|E|}\right) = 15°$

$\beta = \cos^{-1}\left(\frac{E_y}{|E|}\right) = 75°$

5)



dA = Length x width = $y\,dx$

need "y" in terms of x

$y = mx + y\text{-intercept}$ (0)

$y = (\text{slope})x = \left(\frac{b}{a}\right)x$

$dA = y\,dx = \left[\left(\frac{b}{a}\right)x\right]dx$

$Q = \int dq$

$dq = \sigma\,dA = (cx^2)\left(\frac{b}{a}x\,dx\right) = \frac{cb}{a}x^3\,dx$

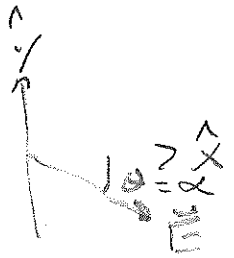
$$Q = \int_0^a dq = \frac{cb}{a}\int_0^a x^3\,dx = \frac{cb}{a}\left[\frac{1}{4}x^4\right]_0^a$$

along x-axis $0 \to a$

$$Q = \frac{cba^4}{4a} = \frac{cba^3}{4}$$

6) c) $\vec{E}$ - field along x-axis

$$\vec{E}\cdot\hat{x} = |\vec{E}|\cos(\alpha) = |\vec{E}|\cos(\theta) = E_x$$



$$E_x = K\lambda\int \frac{\cos(\theta)\,dq}{r^2} \quad \Big| \quad r^2 = y^2 + a^2, \quad \cos\theta = \frac{a}{r} = \frac{a}{(y^2+a^2)^{1/2}}$$

$$E_x = K\lambda\int_0^L \left(\frac{a}{(y^2+a^2)^{1/2}}\right)\left(\frac{1}{(y^2+a^2)}\right)dy$$

$$E_x = Ka\lambda\int_0^L \frac{1}{(y^2+a^2)^{3/2}}\,dy = Ka\left[\frac{y}{a^2(y^2+a^2)^{1/2}}\right]_0^L$$

$$E_x = \frac{K\lambda}{a}\left[\frac{L}{(L^2+a^2)^{1/2}}\right]$$

6 D) $E_y = \vec{E}\cdot\hat{y} = |\vec{E}|\cos(\overbrace{90^\circ+\theta}^{\beta}) = -|\vec{E}|\sin\theta$

$$\sin\theta = \frac{y}{r} = \frac{y}{(y^2+a^2)^{1/2}} \qquad r^2 = (y^2+a^2)$$

$$E_y = -K\lambda\int \frac{\sin\theta\,dq}{r^2} = K\int_0^L \left(\frac{y}{(y^2+a^2)^{1/2}}\right)\left(\frac{1}{(y^2+a^2)}\right)dy$$

$$E_y = -K\lambda\int_0^L \frac{y}{(y^2+a^2)^{3/2}}\,dy = -K\lambda\left[\frac{1}{(y^2+a^2)^{1/2}}\right]_0^L$$

$$E_y = -K\lambda\left[\frac{1}{(L^2+a^2)^{1/2}} - \frac{1}{a}\right]$$

7) $dq = \lambda dL = \lambda R d\theta$

$E_x = \vec{E} \cdot \hat{x} = |\vec{E}| \cos(\alpha) = -|\vec{E}| \cos(\theta)$

$\alpha = 180° - \theta$

$\cos(180° - \theta) = -\cos(\theta)$

$E_x = -\int \frac{k \cos(\theta)\, dq}{R^2} = -\frac{k}{R^2} \int_0^{\pi} \cos(\theta)(\lambda R d\theta)$

$E_x = \frac{-\lambda k}{R} \int_0^{\pi} \cos\theta\, d\theta = -\frac{\lambda k}{R} \left[\sin(\theta)\right]_0^{\pi}$

$E_x = 0 N/c$

$E_y = \vec{E} \cdot \hat{y} = |\vec{E}| \cos(\beta) = -|\vec{E}| \sin(\theta)$

$\beta = 90° + \theta \quad \cos(90° + \theta) = -\sin(\theta)$

$E_y = -\int \frac{k \sin(\theta)}{R^2} dq = -\frac{k}{R^2} \int_0^{\pi} \sin(\theta)\, dq$

$dq = \lambda R d\theta$

$E_y = -\frac{k\lambda}{R} \int_0^{\pi} \sin\theta\, d\theta = \frac{k\lambda}{R} \left[\cos(\theta)\right]_0^{\pi}$

$E_y = \frac{k\lambda}{R}[-1 - 1] = -\frac{2k\lambda}{R}$

$\vec{E} = (0 N/c)\,\hat{x} + \left(-\frac{2k\lambda}{R}\right)\hat{y}$